

The schematic diagram illustrates the battery-side components of a Volthium reader. It includes a power input section at the top left with a connector J1, a fuse F1 (1A 5x20), and a diode D1 (SS24). The main power regulation stage features a TP562933FDRLR (U1) buck converter, which takes input from V24_RAW and provides output to V3V3_SW. This output is filtered by capacitors C1, C2, and C3, and further regulated by a V3V3_CAT5E (U2) LDO. A USB-OTG module (J3) is connected to the V3V3_SW rail. The RTC section includes a CR2032 battery (BAT1) connected to V_BATT_RTC, which is also filtered by capacitor C9. The PSRAM section consists of an ESP32-S3-WROOM-1-N16R8 (MOD1) connected to V3V3_SW and V_BATT_RTC, with various control signals like PWR_EN, ESC_EN, and DBG_QART_TX/RX. The UART section includes a SN65HVD30B2E (U3) transceiver connected to V3V3_SW and V_BATT_RTC, with pins for UART_TX_3V3, UART_RX_3V3, and DBG_QART_TX/RX. The override section includes a BTN_OVERRIDE button connected to V3V3_SW and GND, with a pull-up resistor R13. The final output section includes a TVS2 diode (R5) connected to V3V3_SW and GND, with a pull-up resistor R12. The schematic is organized into several functional blocks, each labeled with its respective component name and value.

Volthium	
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